

yfd8a5w8a

February 17, 2026

```
[1]: import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
from sklearn.metrics import classification_report
import re
import string
```

```
[2]: df_fake = pd.read_csv("Fake.csv")
df_true = pd.read_csv("True.csv")
```

```
[3]: df_fake.head()
```

```
[3]:
```

	title \	text	subject \	date
0	Donald Trump Sends Out Embarrassing New Year'...	Donald Trump just couldn t wish all Americans ...	News	December 31, 2017
1	Drunk Bragging Trump Staffer Started Russian ...	House Intelligence Committee Chairman Devin Nu...	News	December 31, 2017
2	Sheriff David Clarke Becomes An Internet Joke...	On Friday, it was revealed that former Milwauk...	News	December 30, 2017
3	Trump Is So Obsessed He Even Has Obama's Name...	On Christmas day, Donald Trump announced that ...	News	December 29, 2017
4	Pope Francis Just Called Out Donald Trump Dur...	Pope Francis used his annual Christmas Day mes...	News	December 25, 2017

```
[4]: df_true.head(5)
```

```

[4]:                                     title \
0 As U.S. budget fight looms, Republicans flip t...
1 U.S. military to accept transgender recruits o...
2 Senior U.S. Republican senator: 'Let Mr. Muell...
3 FBI Russia probe helped by Australian diplomat...
4 Trump wants Postal Service to charge 'much mor...

                                     text      subject \
0 WASHINGTON (Reuters) - The head of a conservat... politicsNews
1 WASHINGTON (Reuters) - Transgender people will... politicsNews
2 WASHINGTON (Reuters) - The special counsel inv... politicsNews
3 WASHINGTON (Reuters) - Trump campaign adviser ... politicsNews
4 SEATTLE/WASHINGTON (Reuters) - President Donal... politicsNews

                                date
0 December 31, 2017
1 December 29, 2017
2 December 31, 2017
3 December 30, 2017
4 December 29, 2017

```

```

[5]: df_fake["class"] = 0
     df_true["class"] = 1

```

```

[6]: df_fake.shape, df_true.shape

```

```

[6]: ((23481, 5), (21417, 5))

```

```

[7]: # Removing last 10 rows for manual testing
df_fake_manual_testing = df_fake.tail(10)
for i in range(23480,23470,-1):
    df_fake.drop([i], axis = 0, inplace = True)
df_true_manual_testing = df_true.tail(10)
for i in range(21416,21406,-1):
    df_true.drop([i], axis = 0, inplace = True)

```

```

[8]: df_fake.shape, df_true.shape

```

```

[8]: ((23471, 5), (21407, 5))

```

```

[9]: df_fake_manual_testing["class"] = 0
     df_true_manual_testing["class"] = 1

```

C:\Users\zaigh\AppData\Local\Temp\ipykernel_7712\860779283.py:1:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
df_fake_manual_testing["class"] = 0
```

C:\Users\zaigh\AppData\Local\Temp\ipykernel_7712\860779283.py:2:

SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.

Try using `.loc[row_indexer,col_indexer] = value` instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
df_true_manual_testing["class"] = 1
```

```
[10]: df_fake_manual_testing.head(10)
```

```
[10]:
```

		title \
23471	Seven Iranians freed in the prisoner swap have...	
23472	#Hashtag Hell & The Fake Left	
23473	Astroturfing: Journalist Reveals Brainwashing ...	
23474	The New American Century: An Era of Fraud	
23475	Hillary Clinton: 'Israel First' (and no peace ...	
23476	McPain: John McCain Furious That Iran Treated ...	
23477	JUSTICE? Yahoo Settles E-mail Privacy Class-ac...	
23478	Sunnistan: US and Allied 'Safe Zone' Plan to T...	
23479	How to Blow \$700 Million: Al Jazeera America F...	
23480	10 U.S. Navy Sailors Held by Iranian Military ...	

		text	subject \
23471	21st Century Wire says This week, the historic...	Middle-east	
23472	By Dady Chery and Gilbert MercierAll writers ...	Middle-east	
23473	Vic Bishop Waking TimesOur reality is carefull...	Middle-east	
23474	Paul Craig RobertsIn the last years of the 20t...	Middle-east	
23475	Robert Fantina CounterpunchAlthough the United...	Middle-east	
23476	21st Century Wire says As 21WIRE reported earl...	Middle-east	
23477	21st Century Wire says It s a familiar theme. ...	Middle-east	
23478	Patrick Henningsen 21st Century WireRemember ...	Middle-east	
23479	21st Century Wire says Al Jazeera America will...	Middle-east	
23480	21st Century Wire says As 21WIRE predicted in ...	Middle-east	

	date	class
23471	January 20, 2016	0
23472	January 19, 2016	0
23473	January 19, 2016	0
23474	January 19, 2016	0
23475	January 18, 2016	0
23476	January 16, 2016	0
23477	January 16, 2016	0

```

23478  January 15, 2016      0
23479  January 14, 2016      0
23480  January 12, 2016      0

```

```
[11]: df_true_manual_testing.head(10)
```

```

[11]:
                                     title \
21407  Mata Pires, owner of embattled Brazil builder ...
21408  U.S., North Korea clash at U.N. forum over nuc...
21409  U.S., North Korea clash at U.N. arms forum on ...
21410  Headless torso could belong to submarine journ...
21411  North Korea shipments to Syria chemical arms a...
21412  'Fully committed' NATO backs new U.S. approach...
21413  LexisNexis withdrew two products from Chinese ...
21414  Minsk cultural hub becomes haven from authorities
21415  Vatican upbeat on possibility of Pope Francis ...
21416  Indonesia to buy $1.14 billion worth of Russia...

                                     text      subject \
21407  SAO PAULO (Reuters) - Cesar Mata Pires, the ow...  worldnews
21408  GENEVA (Reuters) - North Korea and the United ...  worldnews
21409  GENEVA (Reuters) - North Korea and the United ...  worldnews
21410  COPENHAGEN (Reuters) - Danish police said on T...  worldnews
21411  UNITED NATIONS (Reuters) - Two North Korean sh...  worldnews
21412  BRUSSELS (Reuters) - NATO allies on Tuesday we...  worldnews
21413  LONDON (Reuters) - LexisNexis, a provider of l...  worldnews
21414  MINSK (Reuters) - In the shadow of disused Sov...  worldnews
21415  MOSCOW (Reuters) - Vatican Secretary of State ...  worldnews
21416  JAKARTA (Reuters) - Indonesia will buy 11 Sukh...  worldnews

                                     date  class
21407  August 22, 2017      1
21408  August 22, 2017      1
21409  August 22, 2017      1
21410  August 22, 2017      1
21411  August 21, 2017      1
21412  August 22, 2017      1
21413  August 22, 2017      1
21414  August 22, 2017      1
21415  August 22, 2017      1
21416  August 22, 2017      1

```

```

[12]: df_manual_testing = pd.concat([df_fake_manual_testing,df_true_manual_testing],
    ↪axis = 0)
df_manual_testing.to_csv("manual_testing.csv")

```

```
[13]: df_merge = pd.concat([df_fake, df_true], axis =0 )
df_merge.head(10)
```

```
[13]:
```

	title \	text subject \
0	Donald Trump Sends Out Embarrassing New Year'...	
1	Drunk Bragging Trump Staffer Started Russian ...	
2	Sheriff David Clarke Becomes An Internet Joke...	
3	Trump Is So Obsessed He Even Has Obama's Name...	
4	Pope Francis Just Called Out Donald Trump Dur...	
5	Racist Alabama Cops Brutalize Black Boy While...	
6	Fresh Off The Golf Course, Trump Lashes Out A...	
7	Trump Said Some INSANELY Racist Stuff Inside ...	
8	Former CIA Director Slams Trump Over UN Bully...	
9	WATCH: Brand-New Pro-Trump Ad Features So Muc...	

	date	class
0	December 31, 2017	0
1	December 31, 2017	0
2	December 30, 2017	0
3	December 29, 2017	0
4	December 25, 2017	0
5	December 25, 2017	0
6	December 23, 2017	0
7	December 23, 2017	0
8	December 22, 2017	0
9	December 21, 2017	0

```
[14]: df_merge.columns
```

```
[14]: Index(['title', 'text', 'subject', 'date', 'class'], dtype='object')
```

```
[15]: df = df_merge.drop(["title", "subject", "date"], axis = 1)
```

```
[16]: df.isnull().sum()
```

```
[16]: text      0
      class      0
      dtype: int64
```

```
[17]: df = df.sample(frac = 1)
```

```
[18]: df.head()
```

```
[18]:
```

	text	class
8931	While it s often difficult to notice, Joe Scar...	0
177	It s been a well-known fact for quite some tim...	0
9123	While we believe the UN headquarters would mak...	0
12544	BERLIN (Reuters) - German Chancellor Angela Me...	1
7473	ISTANBUL (Reuters) - Turkish Prime Minister Bi...	1

```
[19]: df.reset_index(inplace = True)
      df.drop(["index"], axis = 1, inplace = True)
```

```
[20]: df.columns
```

```
[20]: Index(['text', 'class'], dtype='object')
```

```
[21]: df.head()
```

```
[21]:
```

	text	class
0	While it s often difficult to notice, Joe Scar...	0
1	It s been a well-known fact for quite some tim...	0
2	While we believe the UN headquarters would mak...	0
3	BERLIN (Reuters) - German Chancellor Angela Me...	1
4	ISTANBUL (Reuters) - Turkish Prime Minister Bi...	1

```
[22]: def wordopt(text):
      text = text.lower()
      text = re.sub('\[.*?\]', '', text)
      text = re.sub("\W", " ", text)
      text = re.sub('https?://\S+|www\.\S+', '', text)
      text = re.sub('<.*?>+', '', text)
      text = re.sub('[%s]' % re.escape(string.punctuation), '', text)
      text = re.sub('\n', '', text)
      text = re.sub('\w*\d\w*', '', text)
      return text
```

```
<>:3: SyntaxWarning: invalid escape sequence '\['
<>:5: SyntaxWarning: invalid escape sequence '\S'
<>:9: SyntaxWarning: invalid escape sequence '\w'
<>:3: SyntaxWarning: invalid escape sequence '\['
<>:5: SyntaxWarning: invalid escape sequence '\S'
```

```
<>:9: SyntaxWarning: invalid escape sequence '\w'
C:\Users\zaigh\AppData\Local\Temp\ipykernel_7712\3432647204.py:3: SyntaxWarning:
invalid escape sequence '\['
    text = re.sub('\[.*?\]', '', text)
C:\Users\zaigh\AppData\Local\Temp\ipykernel_7712\3432647204.py:5: SyntaxWarning:
invalid escape sequence '\S'
    text = re.sub('https?:\/\/\S+|www\.\S+', '', text)
C:\Users\zaigh\AppData\Local\Temp\ipykernel_7712\3432647204.py:9: SyntaxWarning:
invalid escape sequence '\w'
    text = re.sub('\w*\d\w*', '', text)
```

```
[23]: df["text"] = df["text"].apply(wordopt)
```

```
[24]: x = df["text"]
      y = df["class"]
```

```
[25]: x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.25)
```

```
[26]: from sklearn.feature_extraction.text import TfidfVectorizer
      vectorization = TfidfVectorizer()
      xv_train = vectorization.fit_transform(x_train)
      xv_test = vectorization.transform(x_test)
```

```
[27]: from sklearn.linear_model import LogisticRegression
      LR = LogisticRegression()
      LR.fit(xv_train, y_train)
```

```
[27]: LogisticRegression()
```

```
[28]: pred_lr=LR.predict(xv_test)
```

```
[29]: LR.score(xv_test, y_test)
```

```
[29]: 0.9867201426024955
```

```
[30]: print(classification_report(y_test, pred_lr))
```

	precision	recall	f1-score	support
0	0.99	0.99	0.99	5856
1	0.98	0.99	0.99	5364
accuracy			0.99	11220
macro avg	0.99	0.99	0.99	11220
weighted avg	0.99	0.99	0.99	11220

```
[31]: from sklearn.tree import DecisionTreeClassifier
      DT = DecisionTreeClassifier()
      DT.fit(xv_train, y_train)
```

```
[31]: DecisionTreeClassifier()
```

```
[32]: pred_dt = DT.predict(xv_test)
```

```
[33]: DT.score(xv_test, y_test)
```

```
[33]: 0.9946524064171123
```

```
[34]: print(classification_report(y_test, pred_dt))
```

	precision	recall	f1-score	support
0	0.99	1.00	0.99	5856
1	1.00	0.99	0.99	5364
accuracy			0.99	11220
macro avg	0.99	0.99	0.99	11220
weighted avg	0.99	0.99	0.99	11220

```
[35]: from sklearn.ensemble import GradientBoostingClassifier
      GBC = GradientBoostingClassifier(random_state=0)
      GBC.fit(xv_train, y_train)
```

```
[35]: GradientBoostingClassifier(random_state=0)
```

```
[36]: pred_gbc = GBC.predict(xv_test)
```

```
[37]: GBC.score(xv_test, y_test)
```

```
[37]: 0.9953654188948307
```

```
[38]: print(classification_report(y_test, pred_gbc))
```

	precision	recall	f1-score	support
0	1.00	0.99	1.00	5856
1	0.99	1.00	1.00	5364
accuracy			1.00	11220
macro avg	1.00	1.00	1.00	11220
weighted avg	1.00	1.00	1.00	11220

```
[39]: from sklearn.ensemble import RandomForestClassifier
RFC = RandomForestClassifier(random_state=0)
RFC.fit(xv_train, y_train)
```

```
[39]: RandomForestClassifier(random_state=0)
```

```
[40]: pred_rfc = RFC.predict(xv_test)
```

```
[41]: RFC.score(xv_test, y_test)
```

```
[41]: 0.9875222816399287
```

```
[42]: print(classification_report(y_test, pred_rfc))
```

	precision	recall	f1-score	support
0	0.99	0.99	0.99	5856
1	0.99	0.99	0.99	5364
accuracy			0.99	11220
macro avg	0.99	0.99	0.99	11220
weighted avg	0.99	0.99	0.99	11220

```
[43]: def output_lable(n):
    if n == 0:
        return "Fake News"
    elif n == 1:
        return "Not A Fake News"
def manual_testing(news):
    testing_news = {"text":[news]}
    new_def_test = pd.DataFrame(testing_news)
    new_def_test["text"] = new_def_test["text"].apply(wordopt)
    new_x_test = new_def_test["text"]
    new_xv_test = vectorization.transform(new_x_test)
    pred_LR = LR.predict(new_xv_test)
    pred_DT = DT.predict(new_xv_test)
    pred_GBC = GBC.predict(new_xv_test)
    pred_RFC = RFC.predict(new_xv_test)
    return print("\n\nLR Prediction: {} \nDT Prediction: {} \nGBC Prediction: {}
↪ {} \nRFC Prediction: {}".format(output_lable(pred_LR[0]),
↪
↪     output_lable(pred_DT[0]),
↪
↪     output_lable(pred_GBC[0]),
↪     output_lable(pred_RFC[0])))
```

```
[44]: news = str(input())  
      manual_testing(news)
```

```
LR Prediction: Fake News  
DT Prediction: Fake News  
GBC Prediction: Fake News  
RFC Prediction: Fake News
```